Titanic Exploratory Data Analysis (EDA)

Project Overview

This project performs Exploratory Data Analysis (EDA) on the Titanic Dataset using Python and Google Colab. The aim is to explore the dataset, clean missing values, perform statistical analysis, visualize patterns, detect outliers, and identify factors affecting passenger survival.

Objectives

- Explore and understand the dataset structure
- Identify missing values and data quality issues
- Generate descriptive statistics
- Visualize data using different plots
- Detect anomalies and outliers
- Analyze feature correlations
- Test survival-related hypotheses
- Generate meaningful insights and recommendations

Tools and Technologies

- Python
- Google Colab
- Pandas
- NumPy
- Matplotlib
- Seaborn
- SciPy

Dataset Features

Feature	Description
PassengerId	Unique Passenger ID
Survived	Survival Status (0 = No, 1 = Yes)
Pclass	Passenger Class
Name	Passenger Name
Sex	Gender
Age	Passenger Age
SibSp	Number of Siblings/Spouses Aboard
Parch	Number of Parents/Children Aboard
Ticket	Ticket Number
Fare	Ticket Fare
Cabin	Cabin Number
Embarked	Port of Embarkation

Exploratory Data Analysis Performed

Data Inspection

- Dataset Shape
- Data Types
- Missing Value Analysis

Statistical Analysis

- Mean
- Median
- Standard Deviation
- Skewness
- Kurtosis

Data Visualization

- Histogram
- Count Plot
- Box Plot
- Heatmap
- Pair Plot
- FacetGrid

Outlier Detection

- Z-Score Method
- Interquartile Range (IQR) Method

Correlation Analysis

- Pearson Correlation
- Spearman Correlation

Hypotheses Tested

H1: Higher-class passengers had better survival rates.

Result: Confirmed

H2: Age impacted passenger survival.

Result: Confirmed

Key Findings

- Female passengers had significantly higher survival rates.
- First-class passengers survived more frequently than third-class passengers.
- Passenger age influenced survival probability.
- Fare showed a positive relationship with survival.
- Several fare-related outliers were identified.

Recommendations

- Investigate survival patterns among women in third class.
- Create a Family Size feature using SibSp and Parch.
- Apply predictive modeling techniques.
- Perform advanced feature engineering for deeper insights.

Repository Structure

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Titanic-EDA-Project/
│
├── Titanic_EDA.ipynb
├── titanic.csv
├── README.md
```

Author

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